is consistent with the uranium results but makes ambiguous the interpretation of lead ages of meteorites and of Earth.

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GEOPHYSICS

Palæomagnetism of Hawaiian Lava Flows

PALÆOMAGNETIC investigations of volcanic rocks extruded in various parts of the world during the past several million years have generally revealed a younger sequence of lava flows magnetized nearly parallel to the field of a theoretical geocentric axial dipole, underlain by a sequence of older flows with exactly the opposite direction of remanent magnetization. A 180-degree reversal of the geomagnetic field, occurring near the middle of the Pleistocene epoch, has been inferred by many workers from such results[1–3]. This is a preliminary report of an investigation of 755 oriented samples collected from 152 lava flows on the island of Hawaii, selected to represent as many stratigraphic horizons as possible. (Sampling details are indicated in Table 1.) This work was undertaken because Hawaii's numerous thick sequences of lava flows, previously mapped as Pliocene to Historic by Stearns and Macdonald[4], and afterwards assigned ages ranging from later Tertiary to Recent, by Macdonald and Davis[5], appeared to offer an ideal opportunity to examine the most recent reversal of Earth's field.

Remanent magnetization measurements using a spinner type magnetometer have now been completed on at least one sample from each of the 152 lava flows sampled; flow averages of all measurements completed to date before demagnetization are shown in Figs. 1, 2 and 3. Demagnetization experiments of representative samples in alternating fields up to 800 oersteds (peak) typically reduce the intensity of magnetization by 90 per cent but do not significantly change directions of magnetization. Statistical analyses of the direction of magnetization in the samples, before and after demagnetization, and mineralogical studies are in progress.

The preliminary measurements may be summarized as follows: (1) The historic flows have directions of magnetization closely grouped about the present geomagnetic field direction on Hawaii, which has changed less than 2 degrees since about 1900. (2) None of the 152 flows sampled has reversed magnetic polarity. (3) The scatter in directions of magnetization from flow to flow in a given sequence is much smaller than generally reported in comparable studies elsewhere. (4) The average magnetization directions of most of the thick sequences sampled appear to differ significantly both from the present geomagnetic field and from the theoretical field of an axial dipole; moreover, the average directions in different sequences are different from each other.

If the geomagnetic field reversal theory is true, then all the flows sampled on Hawaii are post-reversal and hence younger than early Pleistocene. Alternative explanations (*a*) that lavas were not extruded when the field was reversed or were afterwards eroded or buried, or (*b*) that self-reversal occurred only in lava flows extruded while Earth's field was reversed, are much less plausible. Moreover, the stratigraphic distribution of the 152 lava flows sampled renders unlikely the possibility that a reversely magnetized sequence is present but was inadvertently missed.

The closely parallel directions of magnetization found throughout these rather thick sequences of flows may be interpreted in two ways: (1) An entire sequence may have been extruded in a time-interval too short for large amounts of secular variation to have occurred. However, rates of secular variation as determined from direct measurements at observatories commonly cause more angular variation in several hundred years than is typically observed in the Hawaiian sequences. Accumulation within several hundred years of sequences of more than 50 clearly distinguishable lava flows with total thicknesses up to 450 ft. appears rather unlikely. (2) That low rates of secular variation prevailed in Hawaii during the time these lava flows were extruded is a more plausible explanation. The results from the Historic flows support this interpretation, and the direct measurements back to about 1900 also show less secular variation than is typical elsewhere in the world.

Table 1. SAMPLING DETAILS OF VOLCANIC SERIES EXAMINED

Geological Unit	Age	Flows sampled	Stratigraphic thickness (ft.)	Description
Historic flows	Historic	9	—	Extruded at known historical dates between 1750 and 1955; members of Kau, Hualalai, and Puna Volcanic Series are included (Fig. 1*a*)
Puna Volcanic Series	Late Pleistocene (?) and Recent	18	373 (I) 64 (II)	From Uwekahuna Bluff on west wall of Kilauea caldera. Group II constitutes the lowest 64 ft. of the section, and is separated from Group I by an erosional unconformity (Fig. 1*b*).
Kahuku Volcanic Series	Pleistocene	29	99 (I) 188 (II) 54 (III)	Stratigraphically, flow Groups I, II and III are progressively lower (Fig. 2*a*)
Hamakua Volcanic Series	Late Tertiary and Quaternary	27	135	From south-east side of Laupahoehoe Canyon. These flows comprise the lower members of the Hamakua Volcanic Series, as described by Stearns and Macdonald (ref. 4) (Fig. 2*b*)
Pololu Volcanic Series	Pliocene (?)	50	453	From Waipio Valley. These flows are from the lower member of the series; the highly weathered upper member was not sampled (Fig. 3*a*)
Ninole Volcanic Series	Pliocene (?)	25	116+	From Kaiholena Ridge. The upper 76 ft. were sampled on the east side of the ridge, the lower 40 ft. on the west; an unknown thickness between these is not exposed (Fig. 3*b*)

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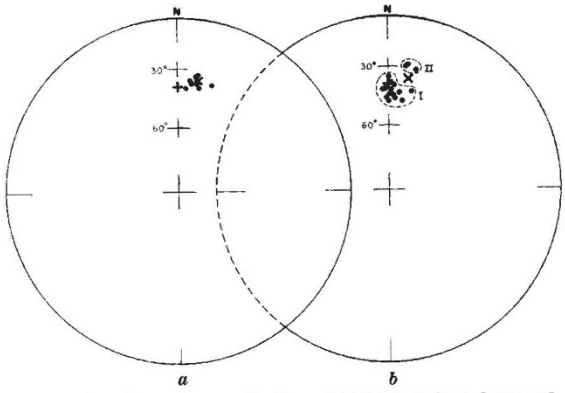


Fig. 1. Directions of magnetization of (*a*) Historic lava flows and (*b*) Puna Volcanic Series. ●, North-seeking polarity downwards; ○, north-seeking polarity upwards; +, direction of axially symmetric dipole field in Hawaii; ×, present geomagnetic field direction in Hawaii. (Lambert equal-area projection with prime circle horizontal)

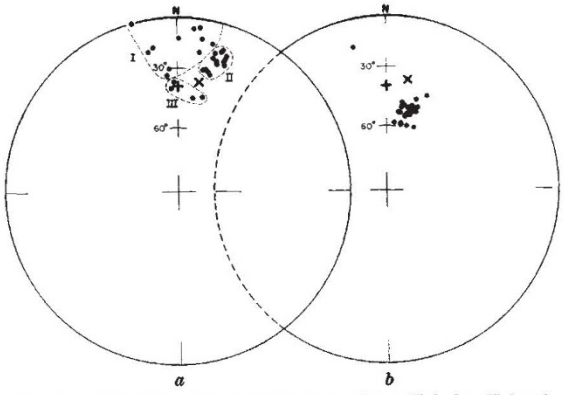


Fig. 2. Directions of magnetization of (*a*) Kahuku Volcanic Series and (*b*) Hamakua Volcanic Series. Symbols as for Fig. 1

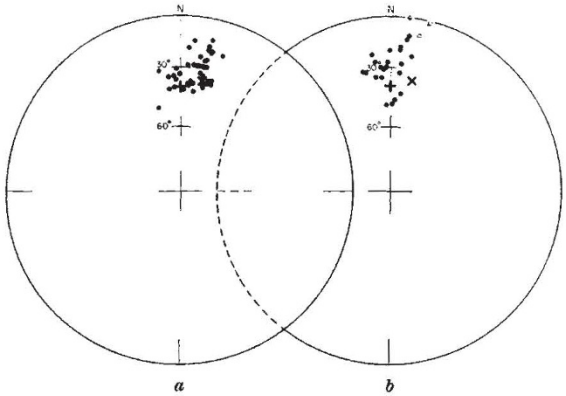


Fig. 3. Directions of magnetization of (*a*) Pololu Volcanic Series and (*b*) Ninole Volcanic Series. Symbols as for Fig. 1

The differences observed between average magnetization directions of the different sequences of flows and the axial dipole field direction do not, in our opinion, indicate that Hawaii has undergone displacements and rotations in the Pleistocene. Such an interpretation would require: (*a*) lower latitudes than at present for Hawaii during the extrusion of the Pololu and Ninole Volcanic Series; (*b*) markedly higher latitudes during the extrusion of the Hamakua Series; (*c*) lower latitudes during Kahuku Series volcanism; (*d*) the present latitude during extrusion of the Puna Series; substantial rotations of the island of Hawaii since the early Pleistocene would also be required. Much more plausible is the explanation that each of these sequences spans a time-interval too short to give an average direction of magnetization parallel to an axial dipole field. Although absolute lengths of time cannot be given, it appears from this investigation that stratigraphic sections 450 ft. thick and composed of more than 50 flows may accumulate on shield volcanoes in periods of time so short that the non-axial dipole field components do not vectorially add to zero. Conversely, the geomagnetic field may have non-axial dipole components for times long enough for the accumulation of these relatively thick sections.

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PHYSICS

Emission of Ultra-Violet and Visible Radiation from Metals by the Impact of High-Energy Hydrogen Ions

In a recent communication[1] we reported that ultra-violet and visible radiation is emitted from nickel by the impact of high-energy protons and positive ions, and that the visible part of the spectrum is continuous. With a resolved proton beam we observed a flat peak between 4600 and 4900 Å. in the distribution curve of spectral intensity.

Apart from the radiation resulting from neutralization of positive ions on metal surfaces[2], the point of most significance is to investigate if the radiation observed by us contains spectral lines or the spectrum of the target element on which high-energy positive ions impinge. If these lines (or spectrum) are present, their origin remains to be traced.

Townes[3] has developed a theory of sputtering of metals under positive ion bombardment based on the assumption of local heating. Kapitza[4] has explained in a similar way the emission of electrons from metal surfaces by positive ions. This theory is supported by the experimental work of Oliphant and others[5], but some theories[6] of sputtering fail to account for recent experimental results. The mechanism of sputtering of metals by positive ions is indeed not yet well understood.

We wish to consider the probable mechanism of the emission of radiation from the target metal. There seem to be two possibilities for the emission of visible and ultra-violet spectra of the target element: (1) the radiation may be emitted from sputtered atoms excited in gaseous phase by positive ions, or by the secondary electrons knocked out of the target; (2) the excited atoms giving out this radiation may originate in the metal itself by the impact

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